SOLUTION

SECTION–I

1. Ans.(B)

$\alpha + \beta = \alpha\beta$

$\Rightarrow \alpha + \beta - \alpha\beta - 1$

$\Rightarrow (\alpha - 1)(1 - \beta) = -1$

$\therefore \alpha = 0 = \beta$ or $\alpha = 2 = \beta$

but $\alpha\beta = \alpha^2 + \beta \Rightarrow \alpha = 0 = \beta$

$\Rightarrow$ only one quadratic equation

2. Ans. (B)

$(x - \alpha)(x - \beta) + (x - \gamma)(x - \delta)$

$= 2(x - m)(x - n)$

$\Rightarrow 2(x - m)(x - n) - (x - \alpha)(x - \beta)$

$= (x - \gamma)(x - \delta)$

So roots are γ, δ.

3. Ans. (D)

$x^2 - 4x - a^2 + 2a + 3 = 0$

given equation

$\Rightarrow (x - 2)^2 = (a - 1)^2$

$\Rightarrow x = 2 \pm (a - 1)$

$\Rightarrow x = 1 + a$ or $3 - a$.

for equation $x^2 - 4x + 4 - a^2 = 0$

$f(1 + a) < 0$ and $f(3 - a) < 0$

$1 + a^2 + 2a - 4 - 4a + 4 - a^2 < 0$

and $9 + a^2 - 6a - 12 + 4a + 4 - a^2 < 0$

$\Rightarrow \frac{1}{2} < a$ and $\frac{1}{2} < a$

$\therefore a \in \left(\frac{1}{2}, 10\right)$

4. Ans. (B)

$\log_3 x = 3 \pm \sqrt{9 - k}$

$\Rightarrow \alpha = 3^{3+\sqrt{9-k}}$ & $\beta = 3^{3-\sqrt{9-k}}$

$\because \frac{\alpha + \beta}{2} \geq \sqrt{\alpha\beta}$

$\Rightarrow \alpha + \beta \geq 2\sqrt{3^6}$

$\alpha + \beta \geq 54$

5. Ans. (B)

$a_n - a_{n-1} = n^2$

Put $n = 2,3,4,....,k$

$\Rightarrow a_k - a_{k-1} = k^2$

$a_{k-1} - a_{k-2} = (k - 1)^2$

:

:

$a_2 - a_1 = 2^2$

Adding, $a_k - a_1 = 2^2 + 3^2 + ...+k^2$

$\Rightarrow 1^2 + 2^2 + k^2 = 204$

$\Rightarrow \frac{k(k+1)(2k+1)}{6} = 204$

$\Rightarrow k = 8$

6. Ans. (B)

Let the first 3 terms be $\frac{a}{r}$, a, ar

Hence $\frac{a}{a\left(r + \frac{1}{r} + 1\right)} = \frac{1}{1 + r + \frac{1}{r}} \leq \frac{1}{3}$ (as $r + \frac{1}{r} \geq 2$)

7. Ans. (C)

Sol. AM $\geq$ GM $\Rightarrow \frac{\frac{12}{x} + \frac{18}{y} + xy}{3} \geq \left(\frac{12}{x} \cdot \frac{18}{y} \cdot xy\right)^{\frac{1}{3}}$

$\Rightarrow \frac{12}{x} + \frac{18}{y} + xy \geq 18$

8. Ans.(C)

$\sum_{i=1}^{6} A_i = 6 \times \left(\frac{2 + 72}{2}\right) = 3 \times 74$

$\prod_{i=1}^{6} G_i = \left(\sqrt{2 \times 72}\right)^6 = 2^3 \times 72^3$

$\sum_{i=1}^{6} \frac{1}{H_i} = 6\frac{\left(\frac{1}{2} + \frac{1}{72}\right)}{2} = \frac{6 \times 74}{4 \times 72}$

$G = 12$

$\Rightarrow \frac{\prod(G_i)\sum\left(\frac{1}{H_i}\right)}{\sum A_i} = 2^2 72^2 = G^4$

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Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

9. Ans. (A,B,D)

Equations are $(ax - b)(x - 1) = 0$

& $(bx - c)(x + 1) = 0$

$\Rightarrow$ Roots are $\frac{b}{a}, 1$ and $\frac{c}{b}, -1$

Case-I : $\frac{b}{a} = \frac{c}{b} \Rightarrow b^2 = ac$

Case-II : $\frac{b}{a} = -1$ & $\frac{c}{b} \neq 1 \Rightarrow -a = b \neq c$

Case-III : $\frac{c}{b} = 1$ & $\frac{b}{a} \neq -1 \Rightarrow c = b \neq -a$

10. Ans. (A,B,D)

Here

$$p(x) = x^6 - x^5 - x^3 - x^2 - x + x^4 - x^4$$

$$= x^2(x^4 - x^3 - x^2) + (x^4 - x^3 - x^2) - x$$

$$= x^2 - x + 1$$

$\therefore p(\alpha) + p(\beta) + p(\gamma) + p(\delta)$

$$= \sum \alpha^2 - \sum \alpha + 4 = (\alpha + \beta + \gamma + \delta)^2$$

$$-2(\alpha\beta + \beta\gamma + \gamma\delta + \alpha\delta + \alpha\gamma + \beta\delta) - \sum \alpha + 4$$

$= 6$

11. Ans. (B,C)

A.M of 1,2,3,.....,n = $\frac{n+1}{2}$

If k, k + 1 are removed then

$$AM = \frac{\frac{n(n+1)}{2} - (2k+1)}{n-2}$$

$$\Rightarrow 1 = \frac{n+1}{2} - \frac{n(n+1) - 2(2k+1)}{2(n-2)}$$

$\Rightarrow n - k = 1$

12. Ans. (B,C)

Sol. Since x, y, z are in A.P., therefore x + z = 2y

Now $\frac{x+y}{2y-x} + \frac{y+z}{2y-z} = \frac{x+y}{z} + \frac{y+z}{x}$

$$= \frac{x}{z} + \frac{y}{z} + \frac{y}{x} + \frac{z}{x}$$

$$= \frac{x}{z} + \frac{x+z}{2z} + \frac{x+z}{2x} + \frac{z}{x} =$$

$1 + \frac{3}{2}\left(\frac{x}{z} + \frac{z}{x}\right) \geq 1 + \frac{3}{2}(2)$ [$\because$ AM $\geq$ G.M.]

$\Rightarrow$ Minimum value of given expression is 4.

Paragraph for Question 13 to 14

13. Ans.(D)

(i) $D \geq 0 \Rightarrow 4((2\lambda + 1)^2 - 2\lambda^2 - 2\lambda) \geq 0$

$\Rightarrow 4(2\lambda^2 + 2\lambda + 1) \geq 0$

true $\forall \lambda \in R$

(ii) $f(-1) > 0$ & $f(1) > 0$

$\Rightarrow \lambda^2 + 5\lambda + 4 > 0$ & $\lambda^2 - 3\lambda > 0$

$\Rightarrow \lambda < -4$ or $\lambda > -1$ & $\lambda < 0$ or $\lambda > 3$

$\Rightarrow \lambda \in (-\infty, -4) \cup (-1, 0) \cup (3, \infty)$

(iii) $-1 < -\frac{b}{2a} < 1 \Rightarrow -1 < \frac{2\lambda + 1}{2} < 1$

$\Rightarrow -\frac{3}{2} < \lambda < \frac{1}{2}$

from (i), (ii) & (iii)

$-1 < \lambda < 0$

$\Rightarrow |10\lambda| \in (0, 10)$

14. Ans.(C)

if both roots are common than

$$\frac{a}{2} = \frac{b}{-2(2\lambda+1)} = \frac{c}{\lambda(\lambda+1)}$$

$\because$ if a, b, c are sides of a triangle than

$a + b > c$

$2 - 2(2\lambda + 1) > \lambda(\lambda + 1)$

$2 - 4\lambda - 2 > \lambda^2 + \lambda$

$\lambda^2 + 5\lambda < 0$

$\lambda \in (-5, 0)$ ...(i)

Similarly when $b + c > a$

$\lambda^2 - 3\lambda - 4 > 0$

$(\lambda - 4)(\lambda + 1) > 0$

$\lambda \in (-\infty, -1) \cup (4, \infty)$ ...(ii)

When $a + c > b$

$\lambda^2 + 5\lambda + 4 > 0$

$\lambda \in (-\infty, -4) \cup (-1, \infty)$ ...(iii)

from (i), (ii) & (iii)

$\lambda \in (-5, -4)$

SECTION–II

1. Ans.5.00

$x^2 + ax + b = 0$ has roots α, β

Replace x by $\frac{1}{x}$ then equation $bx^2 + ax + 1 = 0$

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has roots $\frac{1}{\alpha},\frac{1}{\beta}$ similarly replacing x by $\frac{1}{x}$ in the equation $x^2 - ax + b - 2 = 0$, we get

equation $(b-2)x^2 - ax + 1 = 0$ has roots $\frac{1}{\gamma},\frac{1}{\delta}$

Now, $\alpha\beta = b$, $\gamma\delta = b - 2$ &

$$\frac{1}{\alpha}+\frac{1}{\beta}=-\frac{a}{b},\frac{1}{\gamma}+\frac{1}{\delta}=\frac{a}{b-2}$$

using given conditions we have

$b(b-2) = 24$ & $-\frac{a}{b}+\frac{a}{b-2}=\frac{5}{12}$

$$a\left[\frac{1}{b-2}-\frac{1}{b}\right]=\frac{5}{12}$$

$$a\left[\frac{b-b+2}{24}\right]=\frac{5}{12}$$

$\Rightarrow a = 5$

2. Ans. 1.00

$$f(x)=\frac{x^2-3x+2}{x^2+x-6}$$

$$f(x)=\frac{(x-2)(x-1)}{(x+3)(x-2)}$$

$\because x \neq -3,2$

$$f(x)=\frac{x-1}{x+3}$$

Range : $R-\left\{1,\frac{1}{5}\right\}$

so 1 is the integer which does not lie in the range of $f(x)$

3. Ans. 14.00

$1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + \ldots\ldots\ldots$ up to n terms = 1575

Here two case arise

(i) either n ∈ even (ii) OR n ∈ odd

(i) if n ∈ even.

$$\Rightarrow\left(1^2+3^2+5^2+\ldots\ldots\frac{n}{2}\text{terms}\right)$$

$$+2\left(2^2+4^2+6^2+\ldots\ldots\frac{n}{2}\text{terms}\right)$$

$$\Rightarrow\left(1^2+2^2+3^2+4^2+\ldots\ldots n\text{ terms}\right)$$

$$+4\left(1^2+2^2+3^2+\ldots\ldots\frac{n}{2}\text{terms}\right)$$

$$\Rightarrow\frac{n(n+1)(2n+1)}{6}+\frac{4\left(\frac{n}{2}\right)\left(\frac{n}{2}+1\right)(n+1)}{6}$$

$$\Rightarrow\frac{n(n+1)(2n+1)}{6}+\frac{n(n+2)(n+1)}{6}$$

$$\Rightarrow\frac{n(n+1)}{6}\left[(2n+1)+n+2\right]=\frac{n(n+1)^2}{2}$$

Now if $\frac{n(n+1)^2}{2}=1575$

$\Rightarrow n(n+1)^2 = 2\times1575 = 2\times5\times5\times7\times3\times3 = 14\times(15)^2 \quad \Rightarrow n=14$

(ii) so there is no need to check far n ∈ odd

4. Ans.469.00

Let the three digits be a, ar and ar^2 then number is

$$100a + 10ar + ar^2 \quad \ldots.(i)$$

Given, $a + ar^2 = 2ar + 1$

or $a(r^2 - 2r + 1) = 1$

or $a(r-1)^2 = 1 \quad \ldots.(ii)$

Also given $a + ar = \frac{2}{3}(ar + ar^2)$

$\Rightarrow 3 + 3r = 2r + 2r^2$

$\Rightarrow 2r^2 - r - 3 = 0 \quad \Rightarrow (r+1)(2r-3) = 0$

$\therefore r = -1, 3/2$

for $r = -1$, $a = \frac{1}{(r-1)^2} = \frac{1}{4} \notin I$

$\therefore r \neq -1$

for $r = 3/2$, $a = \frac{1}{\left(\frac{3}{2}-1\right)^2} = 4$

{from (ii)}

From (i), number is $400 + 10.4.\frac{3}{2} + 4.\frac{9}{4} = 469$

Ans.

5. Ans.9.00

$$T_r=\frac{(1+2+3+\ldots+r)(1+3+\ldots+(2r-1))}{r^2(1^3+2^3+\ldots.+r^3)}$$

$$=\frac{\frac{r(r+1)}{2}.r^2}{r^2.\frac{r^2(r+1)^2}{4}}=\frac{2}{r(r+1)}=2\left[\frac{1}{r}-\frac{1}{r+1}\right]$$

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$$\therefore \sum_{r=1}^{10} T_r = 2\left(\frac{1}{1}-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+...+\frac{1}{10}-\frac{1}{11}\right)$$

$$= 2\left(1-\frac{1}{11}\right) = \frac{20}{11}$$

6. Ans. 5.00

$$\underbrace{(1+3+5+....+a)}_{\text{let } n_1 \text{ term}} + \underbrace{(1+3+5+....+b)}_{n_2 \text{ term}} = \underbrace{(1+3+5+....+c)}_{n_3 \text{ term}}$$

$\Rightarrow n_1^2 + n_2^2 = n_3^2$ where

$a = 2n_1 - 1$

$b = 2n_2 - 1$

$c = 2n_3 - 1$

$a + b + c = 57$

$(2n_1 - 1) + (2n_2 - 1) + (2n_3 - 1) = 57$

$2(n_1 + n_2 + n_3) = 60$

$n_1 + n_2 + n_3 = 30$

$\Rightarrow n_1 = 5, n_2 = 12, n_3 = 13$

$\Rightarrow a = 9, b = 23, c = 25$

SECTION–IV

1. Ans. (A)→(P,S); (B)→(P,Q,R,T); (C)→(P,S); (D)→(Q,S)

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(A) $T_q^2 = T_p T_r$

$(aR^{q-1})^2 = aR^{p-1}\ aR^{r-1}$

$2q = p + r$ (A.P.)

(B) If $(a^2+b^2+c^2)(b^2+c^2+d^2) < (ab+bc+cd)^2$

then a,b,c in G.P.

and if $c(a-b) = a(b-c) \Rightarrow$ a,b,c in H.P.

$\Rightarrow$ then a,b,c,d all are equal.

(C) $\frac{a}{b}\log a^2 b,\ \frac{b}{c}\log b^2 c,\ \frac{c}{d}\log c^2 d$

$\frac{1}{r}\log a^3 r,\ \frac{1}{r}\log a^3 r^4,\ \frac{1}{r}\log a^3 r^7$ (A.P.)

(D) $2a_2 = a_1 + a_3$ .....(i)

$a_3^2 = a_2 a_4$ .....(ii)

$a_4 = \frac{2a_3 a_5}{a_3 + a_5}$ .....(iii)

from (i), (ii) & (iii) $\Rightarrow a_3^2 = a_1 a_5$.

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